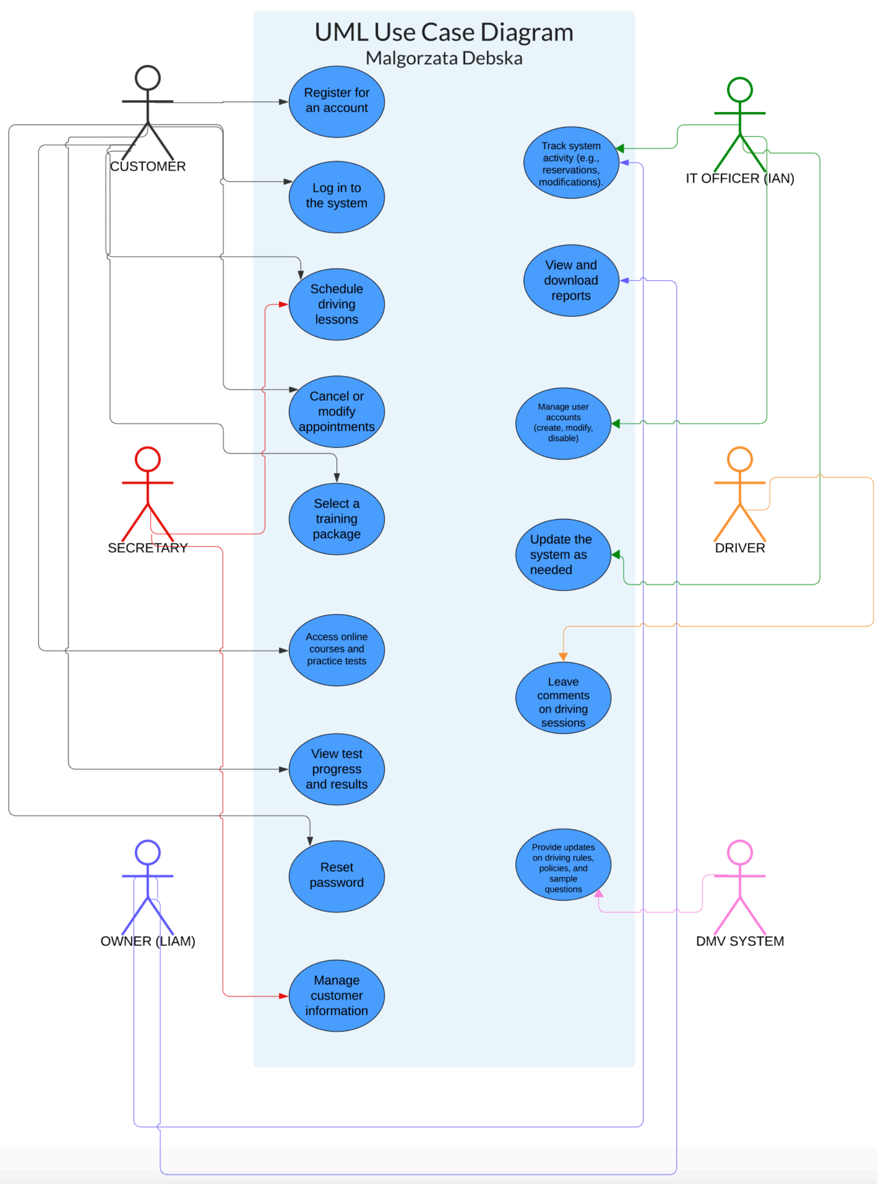
CS 255 System Design Document Template

Malgorzata Debska

This template lays out all the different sections that you need to complete for Project Two. Each section has guidance to prompt your thinking. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead the goal is to complete each section based on what your client's needs are. Remove this note when you are finished, and replace all bracketed text with the relevant information.

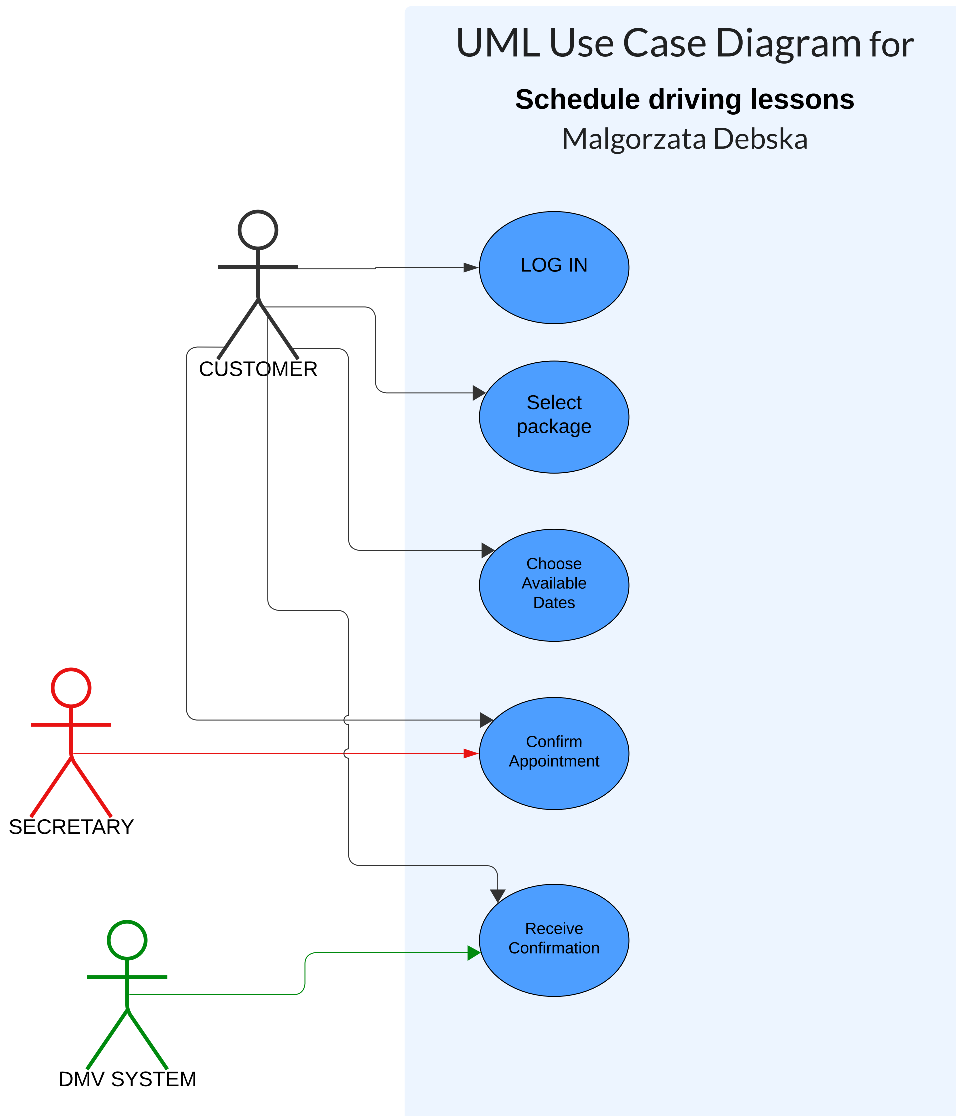
UML Diagrams

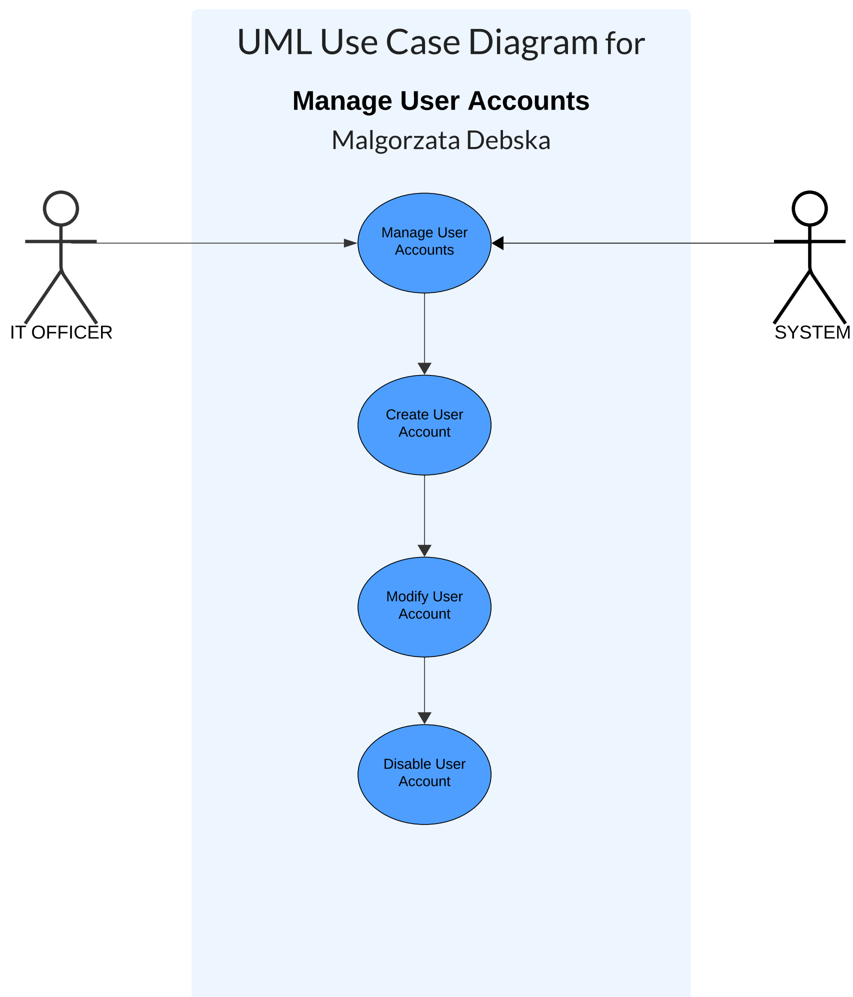
UML Use Case Diagram



Southern New Hampshire University

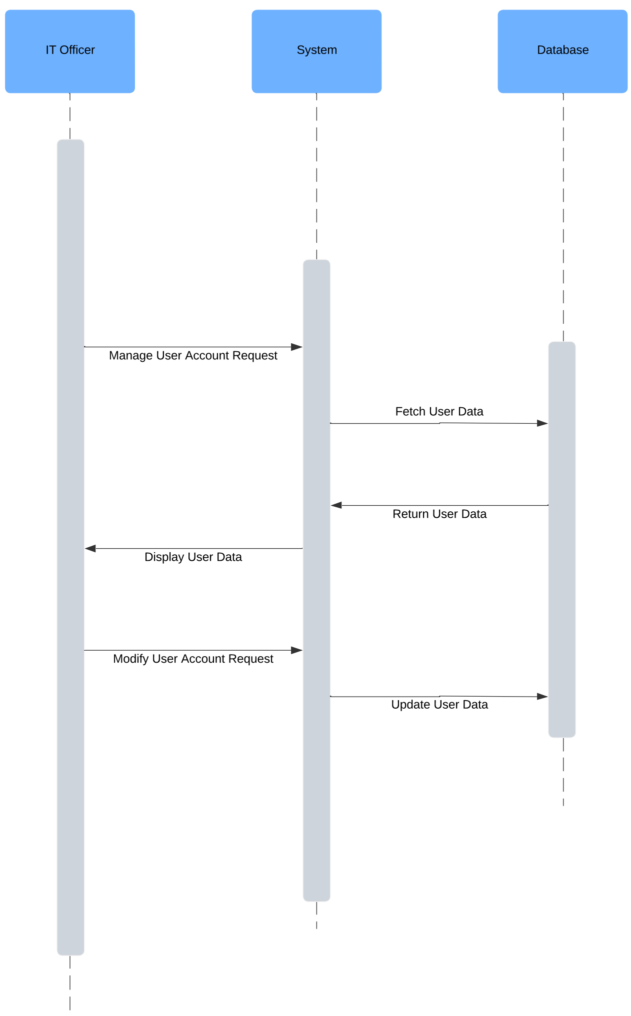
UML Activity Diagrams




UML Use Case Diagram for
Manage User Accounts
Malgorzata Debska
IT OFFICER
SYSTEM
Manage User Accounts
Create User Account
Modify User Account
Disable User Account

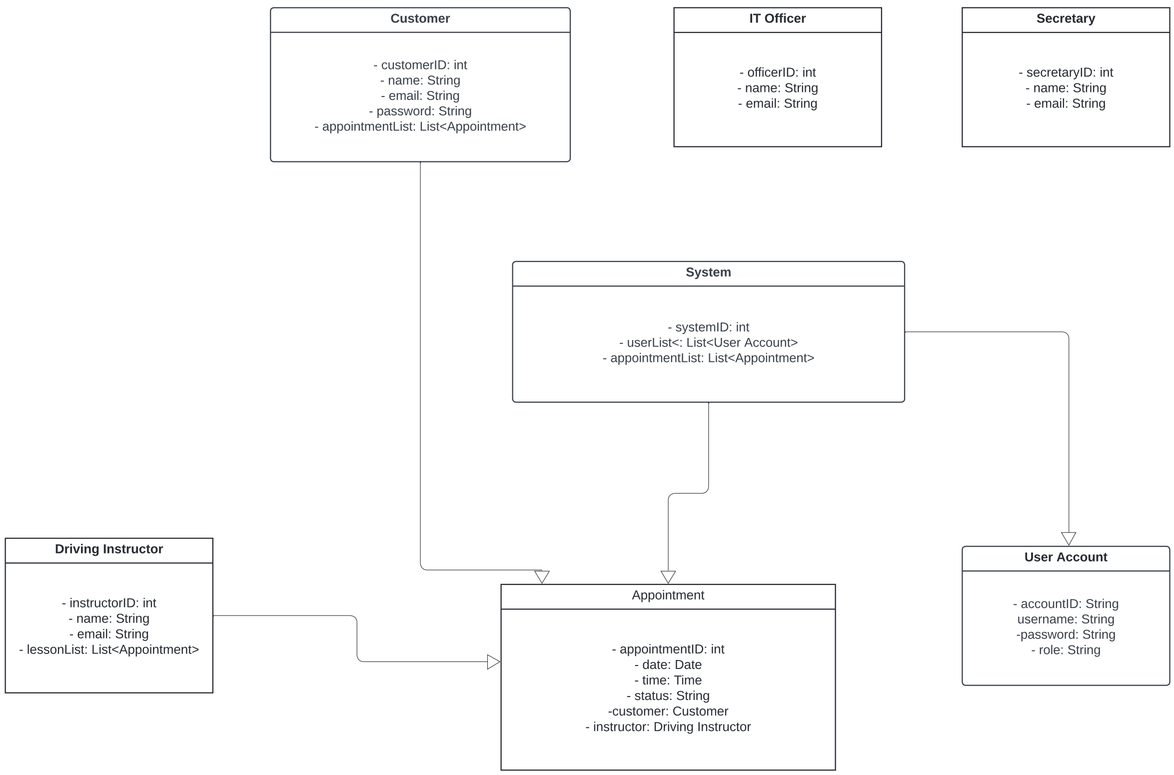
UML Sequence Diagram

SEQUENCE DIAGRAM FOR "MANAGING THE USER ACCOUNT"
Malgorzata Debska



UML Class Diagram

UML CLASS DIAGRAM
Malgorzata Debska



Technical Requirements

The technical requirements for your proposed system are derived from the functional and nonfunctional requirements outlined in your business requirements document. These requirements encompass the necessary hardware, software, tools, and infrastructure to ensure the system operates efficiently, securely, and reliably.

Starting with the **hardware requirements**, the system will need robust servers to host the web application and manage the database. A web server with multi-core processors, 16GB or more of RAM, and SSD storage is essential for handling user interactions and delivering content swiftly. A dedicated database server with similar specifications is required to manage and store critical data, such as user accounts, appointments, and training packages. Additionally, backup and storage devices, like NAS or SAN systems, will be necessary to ensure regular data backups and redundancy. Client devices, including desktops, laptops, and mobile devices, must support modern operating systems to provide users with access to the system from various locations and platforms.

In terms of **software requirements**, the system should be hosted on a Linux-based server operating system, such as Ubuntu Server or CentOS, to leverage their stability and security. The web server software like Apache, will handle HTTP requests and serve content efficiently. The database management system, either MySQL or PostgreSQL, will store and manage the system's data. For development, backend programming languages like Java or Python will be used to implement the business logic, while frontend technologies like HTML, CSS, and JavaScript, along with frameworks such as React or Angular, will ensure a responsive and user-friendly interface. The system will employ SSL/TLS encryption to secure data in transit and use encryption methods for storing sensitive information. Authentication will be managed using OAuth2, supporting multi-factor authentication to enhance security. Version control with Git will facilitate collaborative development and source code management.

The **infrastructure requirements** focus on leveraging cloud-based services to provide scalability, reliability, and ease of maintenance. Cloud providers like AWS, Google Cloud Platform will host the application, database, and storage. Load balancers, such as AWS Elastic Load Balancer, will distribute traffic across multiple servers to ensure high availability. A high-speed internet connection is essential for smooth access, while VPNs will provide secure remote access for IT Admins. Security measures include firewalls to protect against unauthorized access, DDoS protection, and regular security audits to maintain compliance with data protection standards. Automated backup systems will regularly back up the database and application data to a secure off-site or cloud location, supported by a disaster recovery plan to minimize downtime in case of a system failure.

Lastly, the **tools and development environment** will include Integrated Development Environments like Visual Studio Code, IntelliJ IDEA, or PyCharm, tailored to the chosen programming languages. Project management tools such as Jira or Trello will help track progress and manage tasks, while collaboration tools like Slack or Microsoft Teams will facilitate communication. Testing tools, including Selenium for automated web testing and JUnit or pytest for unit testing, will ensure the system functions correctly. Continuous Integration/Continuous Deployment tools like Jenkins will automate the build, test, and deployment processes, allowing for smooth integration of new features into the production environment.